MIL-M-28787/38B(EC)
14 November 1978
SUPERSEDING
MIL-M-28787/38A(EC)
24 March 1976

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, DPDT, POWER, KEY CODE NPM

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic DPDT (double-pole, double-throw) relay module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Relay, DPDT
Family	Power
Key code	NPM
Size	1B
Weight	60 grams maximum
Failure rate	Electrical failure rate
	0.15 failures/10^6 hours maximum
	Mechanical contact failure rate
	0.25 failures/10^6 contact operations maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	2.0 W
Coil voltage and current per relay	8 Vdc at 105 mA maximum
Contact rating	28 Vdc at 2 A maximum
Pick-up voltage	1.4 to 4.5 Vdc
Drop-out voltage	1 to 3.4 Vdc
Operate and release time	18 ms maximum

1.3 Module function. The DPDT relay module, designated as key code NPM, is an electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains four relays each, with two normally-open and two normally-closed contacts. Across each relay coil is a diode for suppression of switching transients.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3.1 Module operating constraints. When operating one or two relays per module simultaneously, the maximum voltage which may be applied to the coil is 8 Vdc per relay. When operating three or four relays per module simultaneously, the maximum coil voltage must be reduced to stay within the module maximum power dissipation of 2.0 watts.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.15 electrical failures per million operating hours and no greater than 0.25 mechanical contact failures per million contact operations when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.4.1 Relay qualification requirements. All relays shall conform to the appendix prior to use in the module.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	Module integrity	II	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B1-B16	Open contact resistance	III	X	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
C1-C16	Closed contact resistance (table IV measured value, see note 4)	IV	X	0.0 (0.0)	0.1 (10.0)	0.0 (0.0)	1.0 (100)	0.0 (0.0)	0.1 (10.0)	0.0 (0.0)	1.0 (100)	Ω (mV)
D1-D4	Coil resistance	V	S	86.0	135.0	86.0	135.0	77.0	153.0	77.0	163.0	Ω
E1-E5	Continuity	VI	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
E6	Continuity, pin 11 to frame	VI	S	0.0	1.0	0.0	1.0	---	---	---	---	Ω
F1-F4	Forward voltage drop	VII	X	0.48	1.0	0.48	1.0	0.48	1.0	0.48	1.0	Vdc
G1-G4	Inverse voltage	VIII	X	0.23	0.39	0.23	0.39	0.21	0.43	0.21	0.43	Vdc
H1-H8	Pick-up voltage	IX	X	1.4	4.0	1.4	4.0	1.4	4.2	1.4	4.5	Vdc
J1-J8	Drop-out voltage	X	X	1.0	3.0	1.0	3.0	1.0	3.2	1.0	3.4	Vdc
K1-K8	Operate time	XI	Q	0	10.0	0	10.0	0	18.0	0	18.0	ms
L1-L8	Release time	XII	Q	0	10.0	0	10.0	0	18.0	0	18.0	ms
M1-M8	Contact bounce (on pick-up)	XIII	Q	0	3.0	0	3.0	0	4.4	0	4.4	ms
M9-M16	Contact bounce (on drop-out)	XIII	Q	0	3.0	0	3.0	0	4.4	0	4.4	ms
N1-N4	Diode inverse current (see 4.5.4)	XIV	Q	0	0.15	0	0.20	0	1.5	0	2.0	μA

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. Test condition C has two sets of limits given. The first set are the limits of the module in the appropriate units; the second set, in parentheses, are the equivalent limits resulting from the test method specified. Table IV gives the equation used in converting from the measured values to the appropriate units.

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3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NPM.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 60 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A20 and B21 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction fin on increment A and heat dissipation fins on increments A and B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/38, except no marking shall interfere with heat dissipation from the fins (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	2
Megohm bridge	Item R06	1
Ohmmeter	Item R01	1
DC voltmeter	Item V04	1
DC ammeter	Item X01	1
Ohmmeter	Item R07	1
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y01	2
DC ammeter	Item X03	1
DC power supply	Item P03	1

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4.3 Test equipment. - Continued

Resistors	No. required (maximum per test figure)
49.9Ω ±1%, 1 W	1
100Ω ±1%, 2 W	1
3.48 kΩ ±1%, 1/10 W	1
28 kΩ ±1%, 1/4 W	1
130Ω ±1%, 1/4 W	1
100 kΩ ±1%, 2 W	1
Transistor	
2N3108, or equivalent	1
Switch	
SPDT	1

All resistors shall be metal film unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence. Test condition N shall be run as specified in 4.5.4.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

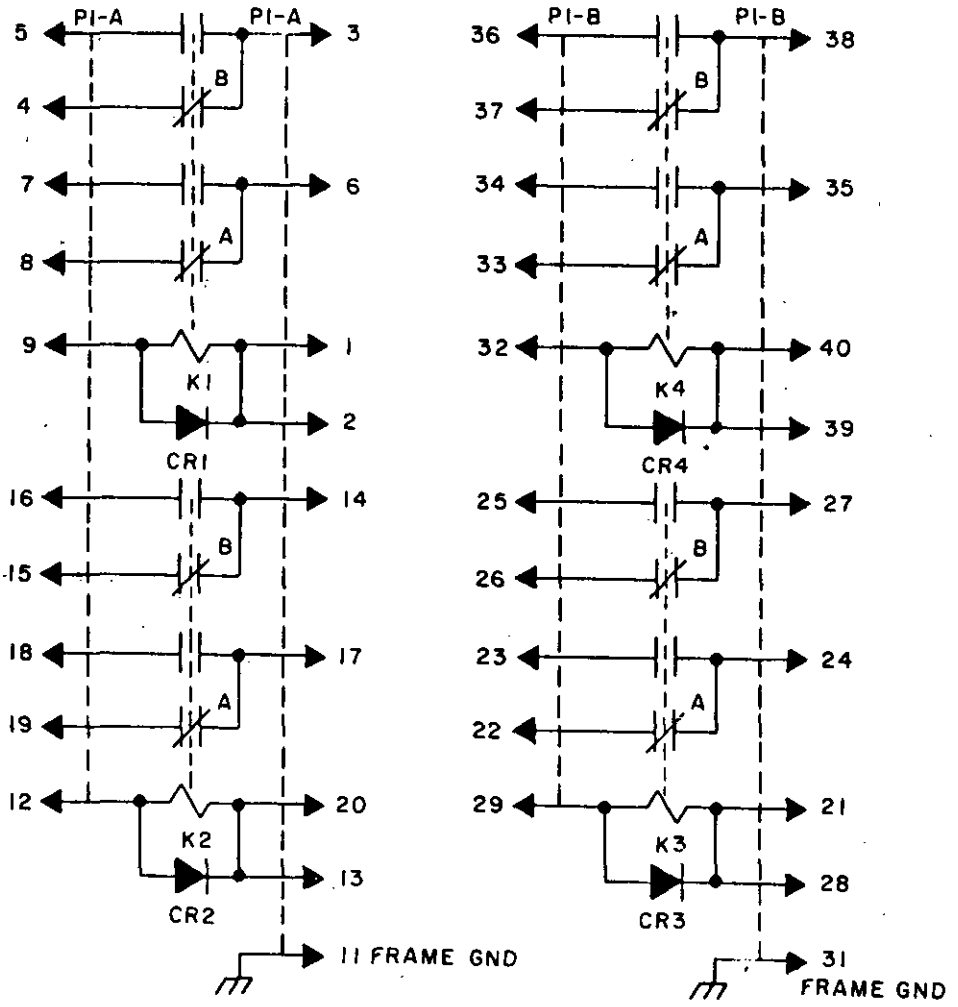
4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through M in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through E5 and F through M shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions A and B shall not be performed at 0°C operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.1.2 Critical component temperature tests. The critical component temperature tests shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, test conditions A through J in table I shall be performed using the initial limits at 25° ±5°C ambient.

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Unused pins: A10 and B30.

FIGURE 1. NPM module function diagram, relay, DPDT.

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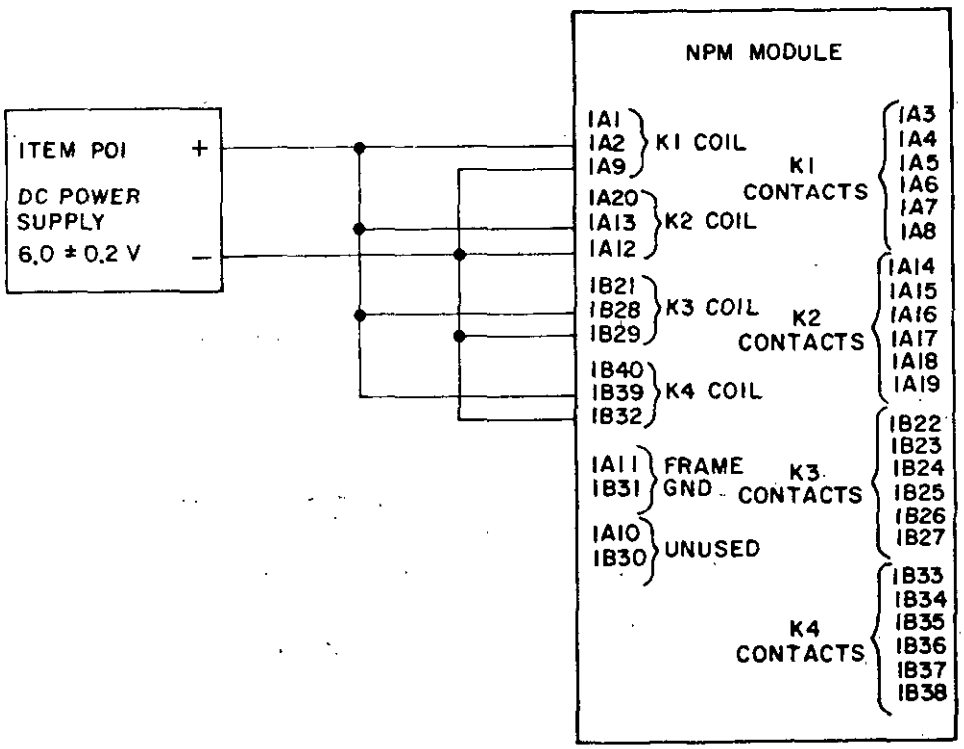
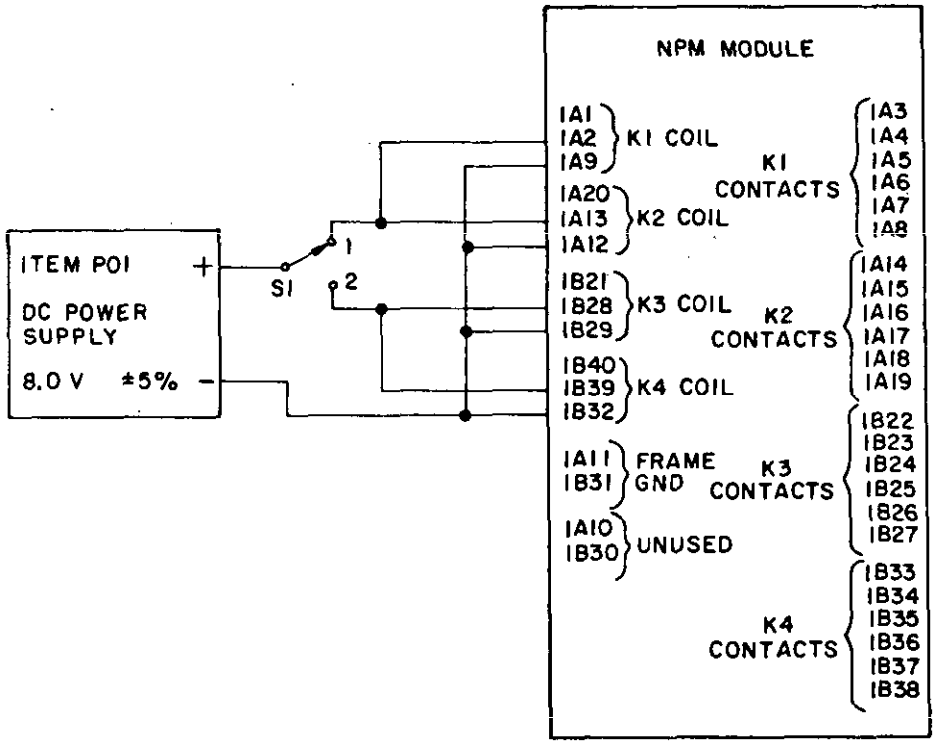


FIGURE 2. Operating temperature and transient thermal test circuit.

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NOTES:
1. For the coil life test, switch S1 shall be set to position 1 for the entire testing period.
2. For the critical component temperature test, switch S1 shall be set to position 1 for one-half the total testing period, then to position 2 for the remainder of the testing period.
3. S1 = SPDT switch.

FIGURE 3. Critical component temperature and coil life-test circuit.

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4.5.3 Coil life test. The coil life test shall be conducted in accordance with MIL-M-28787 life-test requirements. The module shall be operated as shown on figure 3. Following the coil life test, electrical tests A through M shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

4.5.4 Diode inverse current test. The diode inverse current test (test condition N) of table I shall be performed as the last test on one qualification module which has not been subjected to the life test.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

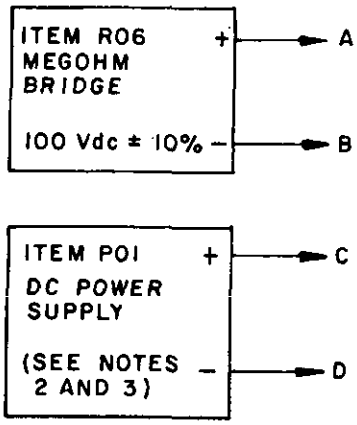
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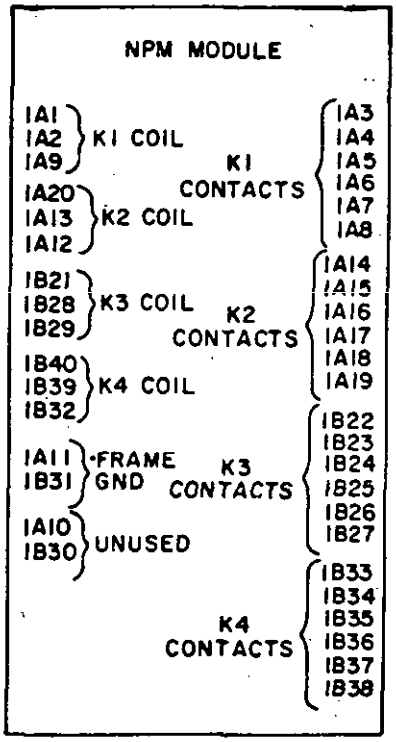
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TABLE II. Module integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Module integrity	1A1-1A9,1A11-1A20, 1B21-1B29,1B31-1B40	1A10, 1B30	10.0	∞	MΩ	4
A2		1A1-1A10,1A12-1A20, 1B21-1B30,1B32-1B40	1A11, 1B31				
A3		1A3, 1A5-1A7	1A9				
A4		1A14, 1A16-1A18	1A12				
A5		1B23-1B25, 1B27	1B29				
A6		1B34-1B36, 1B38	1B32				
A7		1A3, 1A5	1A6, 1A7				
A8		1A14, 1A16	1A17, 1A18				
A9		1B23, 1B24	1B25, 1B27				
A10		1B34, 1B35	1B36, 1B38				
A11		1A1-1A9	1A12-1A20				
A12		1A1-1A9	1B21-1B29				
A13		1A1-1A9	1B32-1B40				
A14		1A12-1A20	1B21-1B29				
A15		1A12-1A20	1B32-1B40				
A16	Module integrity	1B21-1B29	1B32-1B40				
A17	Module integrity (open contact)	1A3, 1A6, 1A14, 1A17, 1B24, 1B27, 1B35, 1B38	1A5, 1A7, 1A16, 1A18, 1B23, 1B25, 1B34, 1B36				
A18	Module integrity (open contact)	1A3, 1A6, 1A14, 1A17, 1B24, 1B27, 1B35, 1B38	1A4, 1A8, 1A15, 1A19, 1B22, 1B26, 1B33, 1B37	10.0	∞	MΩ	4

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. Only during test conditions A17 and A18 connect C to pins 1A2, 1A13, 1B28, and 1B39; connect D to pins 1A9, 1A12, 1B29, and 1B32.
2. For test condition A17 adjust item P01 to 0.8 ±0.1 Vdc.
3. For test condition A18 adjust item P01 to $4.5\ ^{+0.1}_{-0}$ Vdc.

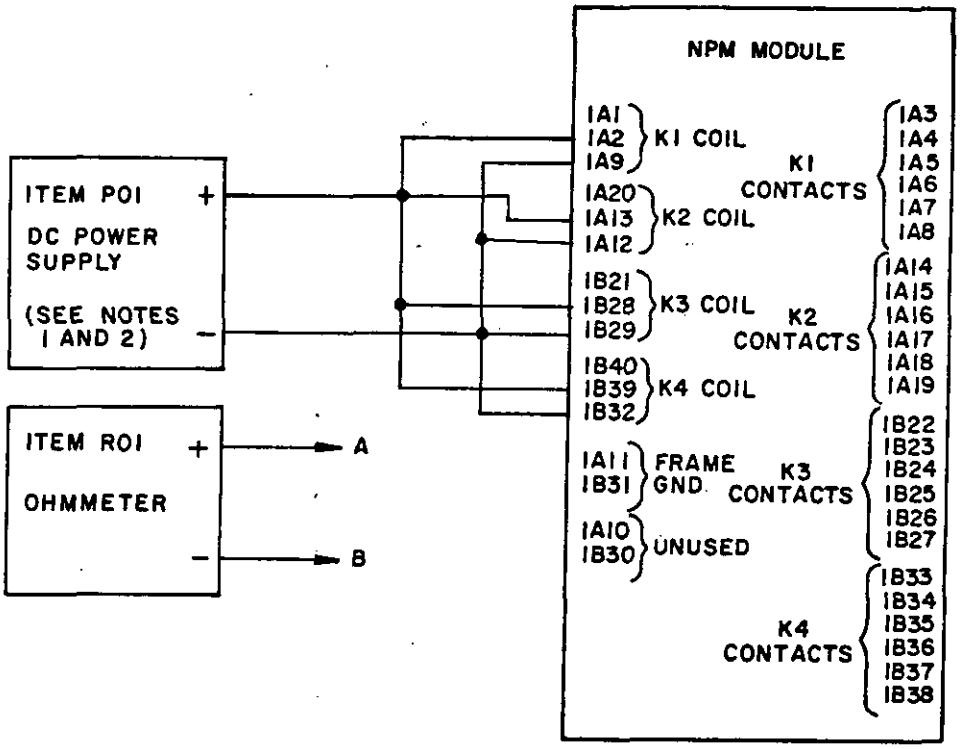
FIGURE 4. Module integrity test circuit for table II.

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TABLE III. Open contact resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Open contact resistance	1A3	1A5	10.0	∞	MΩ	5
B2	↓	1A6	1A7	↓	↓	↓	↓
B3		1A14	1A16				
B4		1A17	1A18				
B5		1B36	1B38				
B6		1B34	1B35				
B7		1B25	1B27				
B8		1B23	1B24				
B9		1A3	1A4				
B10		1A6	1A8				
B11		1A14	1A15				
B12		1A17	1A19				
B13		1B37	1B38				
B14		1B33	1B35				
B15	↓	1B26	1B27	↓	↓	↓	↓
B16	Open contact resistance	1B22	1B24	10.0	∞	MΩ	5

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NOTES:
1. For measurement of test conditions B1-B8, item P01 shall be adjusted to 0.8 ±0.1 Vdc.
2. For measurement of test conditions B9-B16, item P01 shall be adjusted to 4.5 ±0.1 Vdc.

FIGURE 5. Open contact resistance test circuit for table III.

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TABLE IV. Closed contact resistance test.

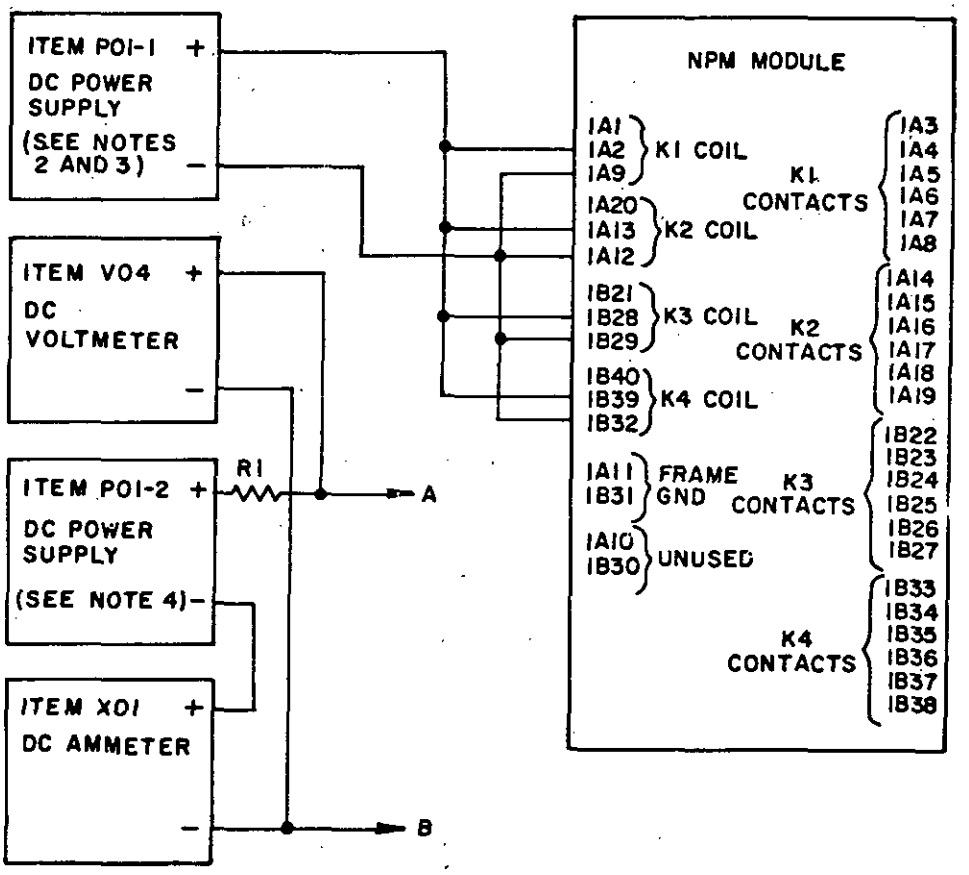
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Closed contact resistance (see note)	1A3	1A4	0	10.0	mV	6
C2	↓	1A6	1A8	↓	↓	↓	↓
C3		1A14	1A15				
C4		1A17	1A19				
C5		1B38	1B37				
C6		1B35	1B33				
C7		1B27	1B26				
C8		1B24	1B22				
C9		1A3	1A5				
C10		1A6	1A7				
C11		1A14	1A16				
C12		1A17	1A18				
C13		1B38	1B36				
C14		1B35	1B34				
C15	↓	1B27	1B25	↓	↓	↓	↓
C16	Closed contact resistance (see note)	1B24	1B23	0	10.0	mV	6

NOTE: The closed contact resistance in ohms can be determined from the following equation:

$$R = \frac{V}{I} = \frac{V}{100}$$

where, V = the measured voltage across the relay contact in mV,
and I = the current through the relay of 100 mA.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 49.9Ω ±1%, 1 W.
2. For test conditions C1-C8 adjust item P01-1 to 0.8 ±0.1 Vdc.
3. For test conditions C9-C16 adjust item P01-1 to 4.5 ±0.1 Vdc.
4. For test conditions C1-C16 adjust item P01-2 for a current of 100 mA as measured on item X01.
5. Voltage sensing leads from the positive and negative terminals of item V04 to the module pins at points A and B shall be connected in such a way as to exclude the resistance (or voltage drop at 100 mA) of the current-carrying leads.

FIGURE 6. Closed contact resistance test circuit for table IV.

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TABLE V. Coil resistance test.

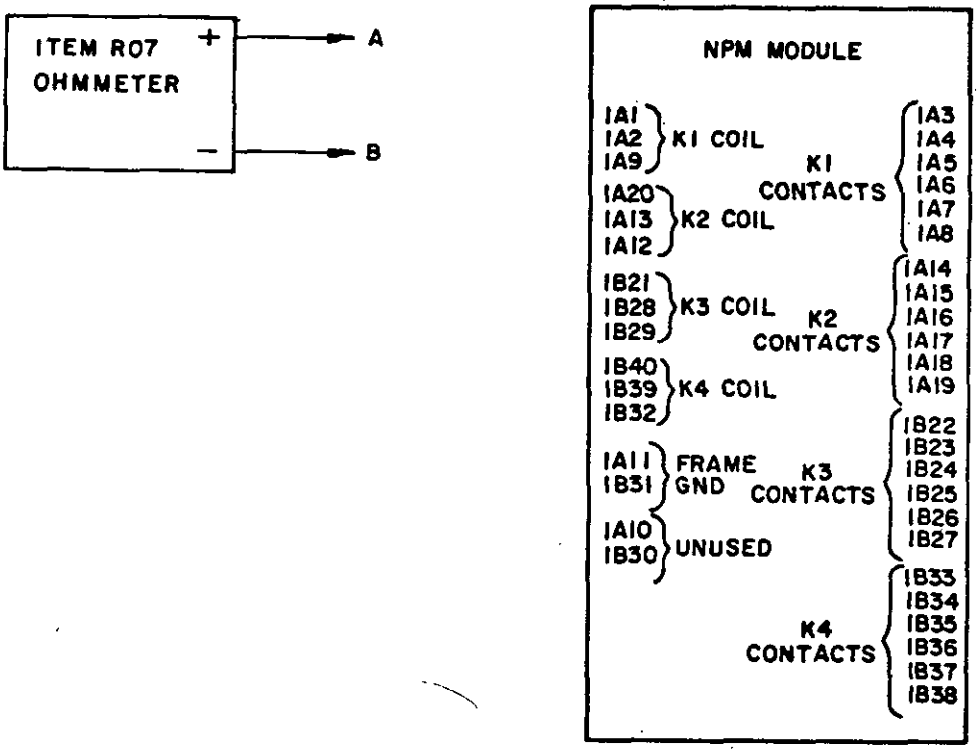
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Coil resistance	1A1	1A9	86.0	135.0	Ω	7
D2	↓	1A20	1A12	↓	↓	↓	↓
D3		1B40	1B32				
D4	Coil resistance	1B21	1B29	86.0	135.0	Ω	7

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TABLE VI. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Continuity	1A1	1A2	0	1.0	Ω	7
E2	↓	1A20	1A13	↓	↓	↓	↓
E3	↓	1B40	1B39	↓	↓	↓	↓
E4	↓	1B21	1B28	↓	↓	↓	↓
E5	↓	1A11	1B31	↓	↓	↓	↓
E6	Continuity	1A11	Frame	0	1.0	Ω	7

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NOTE: For test condition E6 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Coil resistance and continuity test circuit for tables V and VI.

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TABLE VII. Forward voltage drop test.

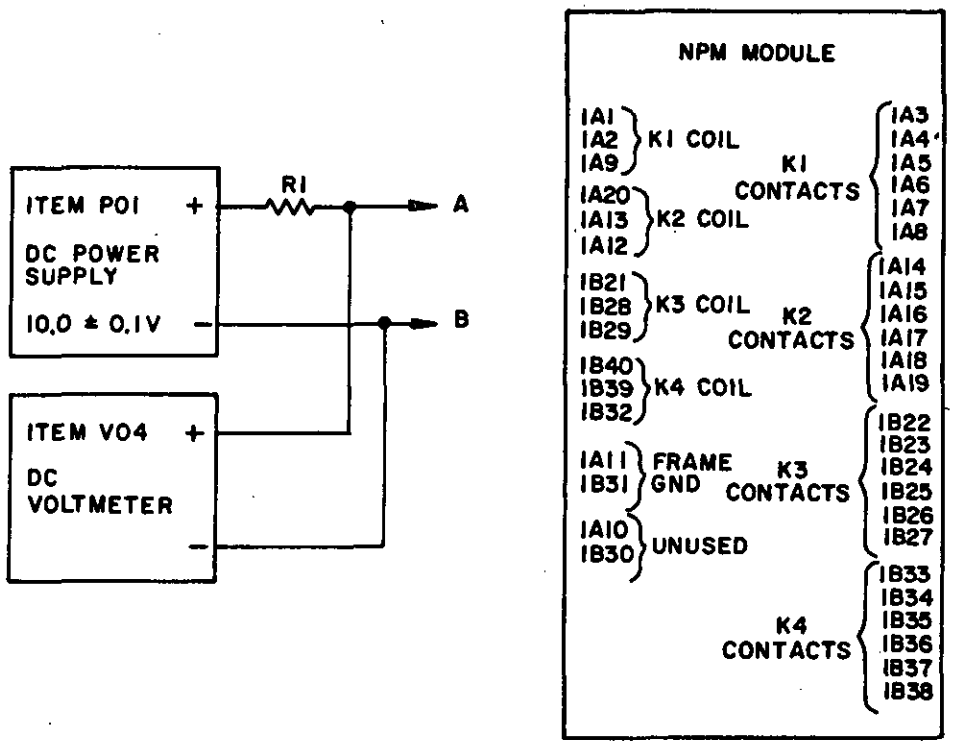
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Forward voltage drop	1A9	1A1	0.48	1.0	Vdc	8
F2	↓	1A12	1A20	↓	↓	↓	↓
F3		1B29	1B21				
F4	Forward voltage drop	1B32	1B40	0.48	1.0	Vdc	8

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TABLE VIII. Inverse voltage test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Inverse voltage	1A1	1A9	0.23	0.39	Vdc	8
G2	↓	1A20	1A12	↓	↓	↓	↓
G3		1B21	1B29				
G4	Inverse voltage	1B40	1B32	0.23	0.39	Vdc	8

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NOTES:
1. For test conditions F1-F4, R1 = 100Ω ±1%, 2 W.
2. For test conditions G1-G4, R1 = 3.48 kΩ ±1%, 1/10 W.

FIGURE 8. Forward voltage and inverse voltage test circuit for tables VII and VIII.

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TABLE IX. Pick-up voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Pick-up voltage	1A2	1A9	1A3	1A5	1.4	4.0	Vdc	9
H2	↓	1A2	1A9	1A6	1A7	↓	↓	↓	↓
H3		1A13	1A12	1A14	1A16				
H4		1A13	1A12	1A17	1A18				
H5		1B28	1B29	1B24	1B23				
H6		1B28	1B29	1B27	1B25				
H7	↓	1B39	1B32	1B35	1B34	↓	↓	↓	↓
H8	Pick-up voltage	1B39	1B32	1B38	1B36	1.4	4.0	Vdc	9

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TABLE X. Drop-out voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	Drop-out voltage	1A2	1A9	1A3	1A4	1.0	3.0	Vdc	9
J2	↓	1A2	1A9	1A6	1A8	↓	↓	↓	↓
J3		1A13	1A12	1A14	1A15				
J4		1A13	1A12	1A17	1A19				
J5		1B28	1B29	1B24	1B22				
J6		1B28	1B29	1B27	1B26				
J7		1B39	1B32	1B35	1B33				
J8	Drop-out voltage	1B39	1B32	1B38	1B37	1.0	3.0	Vdc	9

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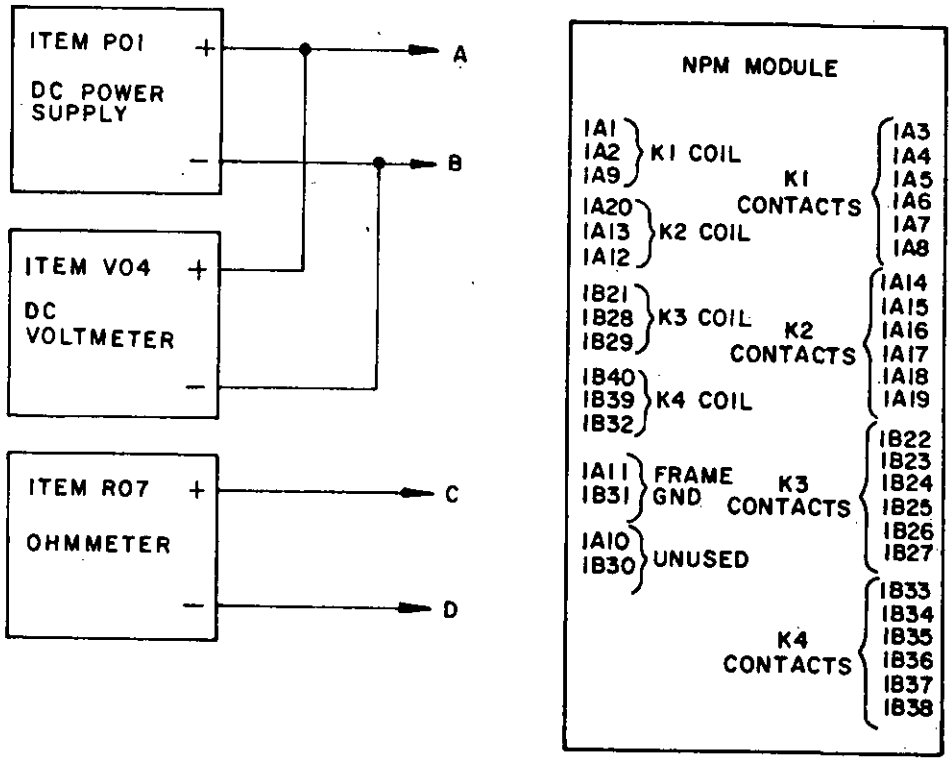


FIGURE 9. Pick-up and drop-out voltage test circuit for tables IX and X.

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NOTES:

1. Test procedure for test conditions H1-H8:

   (a) Connect test points A, B, C, and D as specified for test condition H1.
   (b) Adjust output of item P01 for 8.0 Vdc ±5% as measured on item V04. Maintain this voltage for a period of 1 to 3 seconds.
   (c) Gradually reduce the output voltage of item P01 to zero. The contact resistance, as measured on item R07, should be 1 megohm or greater.
   (d) Gradually increase the output voltage of item P01 until the contact resistance, as measured on item R07, drops to 1 ohm or less.
   (e) The pick-up voltage shall be read on item V04, and shall be as specified for test condition H1.
   (f) Repeat notes (a) through (e) for test conditions H2-H8.

2. Test procedure for test conditions J1-J8:

   (a) Connect test points A, B, C, and D as specified for test condition J1.
   (b) Adjust output of item P01 for 8.0 Vdc ±5% as measured on item V04. The contact resistance, as measured on item R07, should be 1 megohm or greater.
   (c) Gradually reduce the output voltage of item P01 until the contact resistance, as measured on item R07, drops to 1 ohm or less.
   (d) The drop-out voltage shall be read on item V04, and shall be as specified for test condition J1.
   (e) Repeat notes (a) through (d) for test conditions J2-J8.

FIGURE 9. Pick-up voltage and drop-out voltage test circuit for tables IX and X - Continued.

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TABLE XI. Operate time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Operate time	1A3	1A5	0	10.0	ms	10, 11
K2	↓	1A6	1A7	↓	↓	↓	↓
K3	↓	1A14	1A16	↓	↓	↓	↓
K4	↓	1A17	1A18	↓	↓	↓	↓
K5	↓	1B24	1B23	↓	↓	↓	↓
K6	↓	1B27	1B25	↓	↓	↓	↓
K7	↓	1B35	1B34	↓	↓	↓	↓
K8	Operate time	1B38	1B36	0	10.0	ms	10, 11

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TABLE XII. Release time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Release time	1A3	1A4	0	10.0	ms	10, 11
L2	↓	1A6	1A8	↓	↓	↓	↓
L3		1A14	1A15				
L4		1A17	1A19				
L5		1B24	1B22				
L6		1B27	1B26				
L7		1B35	1B33				
L8	Release time	1B38	1B37	0	10.0	ms	10, 11

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TABLE XIII. Contact bounce test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Contact bounce	1A3	1A5	0	3.0	ms	10, 11
M2		1A6	1A7				
M3		1A14	1A16				
M4		1A17	1A18				
M5		1B24	1B23				
M6		1B27	1B25				
M7		1B35	1B34				
M8		1B38	1B36				
M9		1A3	1A4				
M10		1A6	1A8				
M11		1A14	1A15				
M12		1A17	1A19				
M13		1B24	1B22				
M14		1B27	1B26				
M15		1B35	1B33				
M16	Contact bounce	1B38	1B37	0	3.0	ms	10, 11

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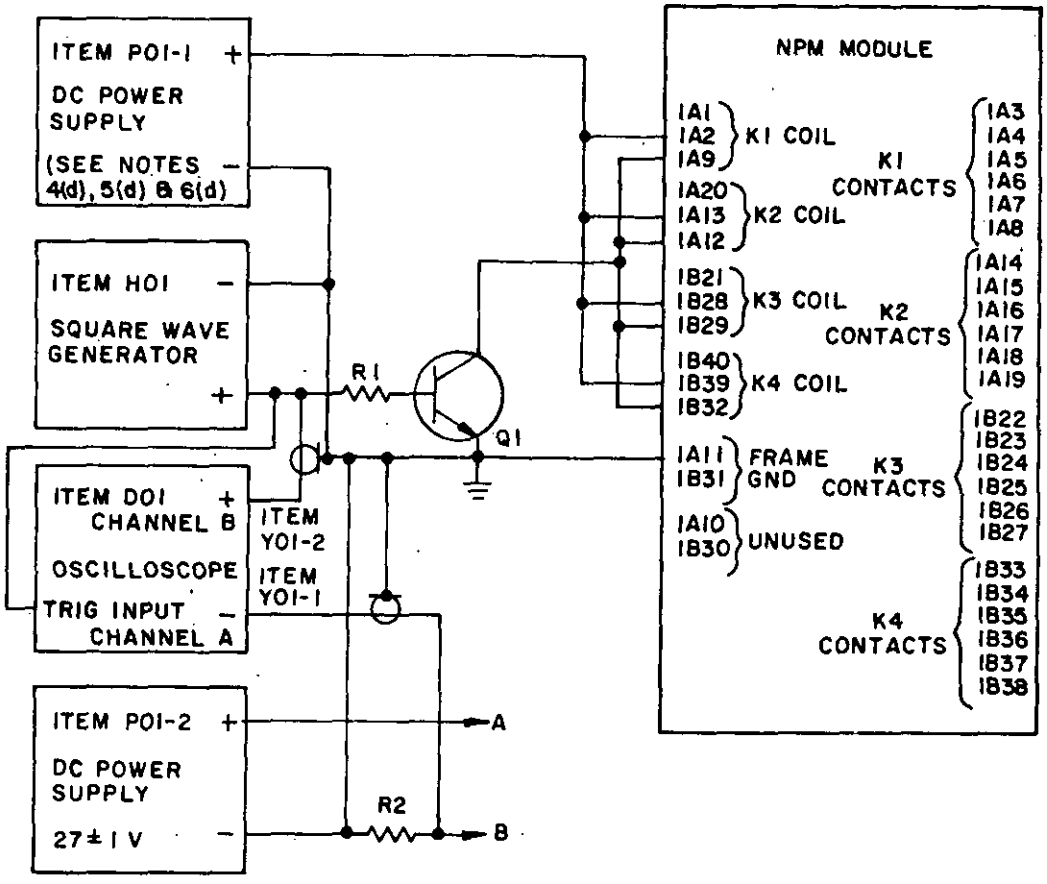


FIGURE 10. Operate time, release time, and contact bounce test circuit for tables XI, XII, and XIII.

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NOTES:

1. R1 = 130Ω ±1%, 1/4 W.
2. R2 = 28 kΩ ±1%, 1/4 W.
3. Q1 = 2N3108, or equivalent.
4. Test procedure for test conditions K1-K8:
   (a) Connect test points A and B as specified for test condition K1.
   (b) On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to positive.
   (c) Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 8.0 ±0.3 Vp-p as read on item D01, channel B.
   (d) Adjust output of item P01-1 to 4.6 ±0.1 Vdc with item H01 set as specified in (c) above. Measure the Q1 "on" collector voltage (V_{CE}). Increase the item P01-1 voltage by the measured value of V_{CE} to 4.6 + V_{CE} ±0.1 Vdc.
   (e) The operate time shall be read on item D01, channel A, and shall be as specified for test condition K1.
   (f) Repeat notes (a) through (e) for test conditions K2-K8.
5. Test procedure for test conditions L1-L8:
   (a) Connect test points A and B as specified for test condition L1.
   (b) On item D01, set the trigger coupling to dc, the trigger source to external, the trigger level to positive, and the trigger slope to negative.
   (c) Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 8.0 ±0.3 Vp-p as read on item D01, channel B.
   (d) Adjust output of item P01-1 to 4.6 ±0.1 Vdc with item H01 set as specified in (c) above. Measure the Q1 "on" collector voltage (V_{CE}). Increase the item P01-1 voltage by the measured value of V_{CE} to 4.6 + V_{CE} ±0.1 Vdc.
   (e) The release time shall be read on item D01, channel A, and shall be as specified for test condition L1.
   (f) Repeat notes (a) through (e) for test conditions L2-L8.
6. Test procedure for test conditions M1-M16:
   (a) Connect test points A and B as specified for test condition M1.
   (b) On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to positive for test conditions M1-M8. The trigger slope shall be set to negative for test conditions M9-M16.
   (c) Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 8.0 ±0.3 Vp-p as read on item D01, channel B.
   (d) Adjust output of item P01-1 to 4.6 ±0.1 Vdc with item H01 set as specified in (c) above. Measure the Q1 "on" collector voltage (V_{CE}). Increase the item P01-1 voltage by the measured value of V_{CE} to 4.6 + V_{CE} ±0.1 Vdc.
   (e) The contact bounce shall be read on item D01, channel A and shall be as specified for test condition M1.
   (f) Repeat notes (a) through (e) for test conditions M2-M16.

FIGURE 10. Operate time, release time, and contact bounce test circuit for tables XI, XII, and XIII - Continued.

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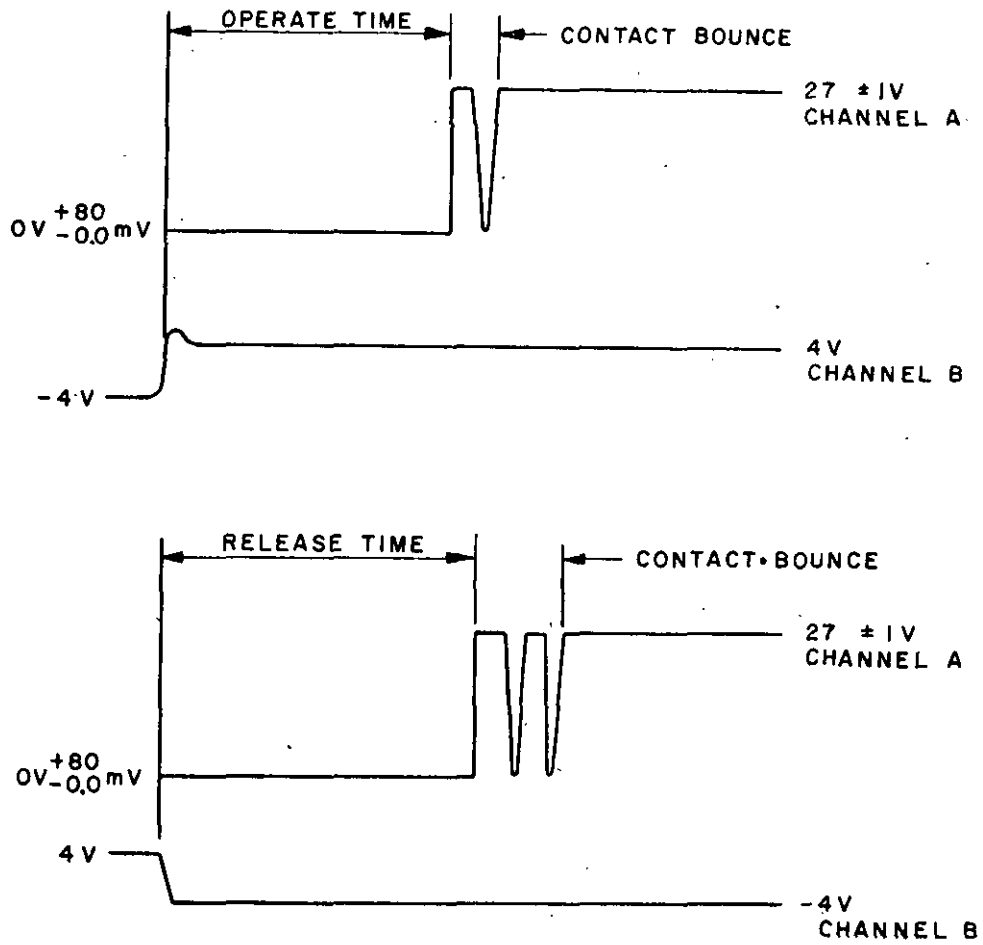


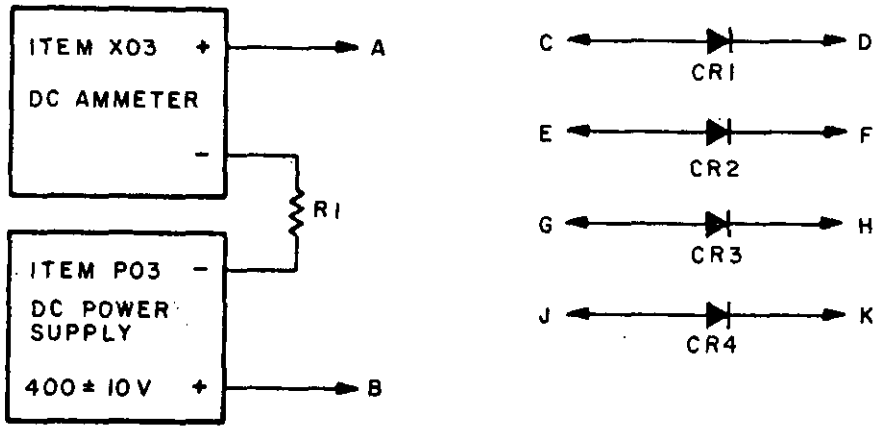
FIGURE 11. Typical oscilloscope traces used to determine the operate time, release time, and contact bounce for tables XI, XII, and XIII.

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TABLE XIV. Diode inverse current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Diode inverse current	C	D	0	0.15	µA	12
N2	↓	E	F	↓	↓	↓	↓
N3	↓	G	H	↓	↓	↓	↓
N4	Diode inverse current	J	K	0	0.15	µA	12

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WARNING:

High voltage (400 V) will be present at the output binding posts of item P03 during test conditions N1-N4. Always reduce the output voltage to zero volts before connecting or disconnecting test leads.

NOTES:
1. R1 = 100 kΩ ±1%, 2 W.
2. Diodes CR1 through CR4 shall be removed from the module for test conditions N1-N4.

FIGURE 12. Diode inverse current test circuit for table XIV.

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APPENDIX

DOUBLE-POLE, DOUBLE-THROW RELAY

10. SCOPE

10.1 Scope. This appendix establishes the requirements and test procedures for a sensitive, double-pole, double-throw (DPDT) relay.

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this appendix to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-5757 - Relays [Electrical (excluding Thermal) for Electronic and Communication Type Equipment], General Specification for.

MIL-Q-9858 - Quality Program Requirements.

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of U. S. Military Property.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

PUBLICATIONS

MILITARY

OP 400 - General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

OD 37626 - Relay Inspection Handbook.

(All Ordnance Data publications shall be ordered from the Commanding Officer, Technical Documents Department, Naval Ordnance Station, Louisville, KY 40214.)

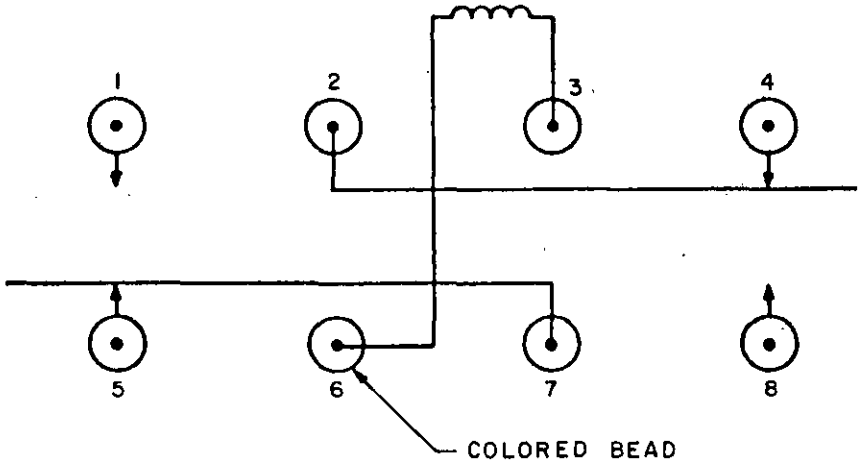
30. REQUIREMENTS

30.1 General requirements. The relay shall comply with the requirements of MIL-R-5757, enclosure 4, vibration characteristic 1, temperature class A, except as modified herein. The schematic shall be as shown on figure 13.

30.2 Contacts. The contact arrangement shall be two-pole, double-throw (2 Form C) with an electrical break-before-make at a contact load of 0.050 A resistive with 28 Vdc across the fixed contacts.

30.3 Electrical characteristics. The relay shall have the following electrical characteristics.

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NOTE: Numbers are for reference only.

FIGURE 13. DPDT relay schematic diagram.

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30.3.1 Life and rated load requirements. The relay shall be capable of operating satisfactorily at the following loads for the designated operating cycles. Relays shall be capable of switching dc contact loads at either contact polarity.

a. For 100,000 cycles
   1. 2A, 28 Vdc resistive switching
   2. 0.2A, 28 Vdc inductive switching
   3. 0.25A, 28 Vdc resistive switching
   4. 0.100A, 28 Vdc resistive switching

b. For 1,000,000 cycles
   1. 0.060A, 28 Vdc resistive switching
   2. 10 µA, 10 mV, ac or dc resistive switching

c. For 200 cycles (relay case ungrounded)
   1. Make 8A, 0.9 to 0.95 power factor, 115 Vac, 400 Hz, inductive load (the 8A has 5 ms duration, then decays to 1A). Break 1A, 0.9 to 0.95 power factor, 115 Vac, 400 Hz.

30.3.2 Static contact characteristics. The static contact characteristics shall be determined using the circuit configuration specified on figure 14 and shall be as follows:

a. The voltage drop across the closed contacts shall be 50 µVdc maximum at a contact current (I) of 1 mAdc ±3%. The open circuit voltage (V) across the open contacts shall be from 10 to 50 mVdc. The closed contact voltage reading following life test 3 shall be 100 µVdc maximum.

b. The resistance of the closed contacts shall be 0.050 ohm maximum, I from 1 to 100 mAdc before life tests.

c. The resistance of the closed contacts shall be 0.100 ohm maximum, I from 1 to 100 mAdc after start of life tests, except the closed contact resistance shall be 1 ohm maximum after start of life test 2.

30.3.3 Electrical requirements. The relay electrical requirements shall be as follows:

a. Unless otherwise specified, all electrical requirements shall apply from 0° to 85°C and acceptance tests shall conform to table XV.

b. The operating voltage range of the relay shall be 4.2 to 11.0 Vdc from 0° to 50°C and 4.8 to 11.0 Vdc from above 50° to 85°C.

c. The relay coil resistance shall be 89 to 132 ohms at 25° ±5°C and 3 mA maximum coil current. For temperatures other than 25°C, the coil resistance shall follow the standard copper wire cure within ±5 percent.

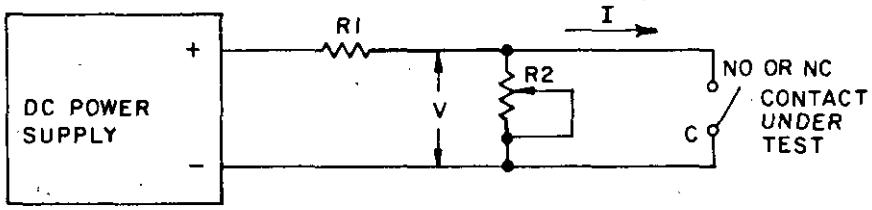
d. The relay pick-up voltage range shall be 1.4 to 4.0 Vdc from 0° to 50°C, 1.4 to 4.2 Vdc from above 50° to 60°C, and 1.4 to 4.5 Vdc from above 60° to 85°C.

e. The relay drop-out voltage range shall be 1 to 3 Vdc from 0° to 50°C, 1.0 to 3.2 Vdc from above 50° to 60°C, and 1.0 to 3.4 Vdc from above 60° to 85°C.

f. The operate and release times of the relay shall each be 10 ms maximum from 0° to 50°C and 18 ms maximum from above 50° to 85°C.

g. The contact bounce of the relay shall be 3 ms maximum on pick-up or drop-out from 0° to 50°C and 4.4 ms maximum on pick-up or drop-out from above 50° to 85°C.

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NOTE: The dc power supply and resistances R1 and R2 shall be selected to obtain a contact current (I) of 1 mAdc ±3% (contact closed) and an open circuit voltage (V) of 10 to 50 mVdc (contact open).

FIGURE 14. Static contact characteristic test circuit.

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30.4 Overvoltage. The relay release time shall not exceed 20 ms when tested in accordance with the overvoltage conditions specified in 40.7.7.

30.5 Overvoltage protection. A silicon rectifier, type 1N647 or equivalent, shall be connected across the coil terminals external to the relay can for all tests (rectifier not to be shipped with relay).

30.6 Run-in test. Each relay shall be subjected to the 5,000 cycle run-in test by the manufacturer, prior to shipment, in accordance with 40.7.2. When tested in accordance with the conditions specified in 40.7.2, the relay contact resistance shall not exceed 1 ohm (the voltage drop shall not exceed 1 millivolt) for a closed contact during any cycle.

30.7 Insulation resistance and dielectric strength. The relay insulation resistance and dielectric strength requirement shall be as follows.

30.7.1 Insulation resistance. The insulation resistance shall be as follows:

a. Before life tests: 5,000 megohms minimum, except that the insulation resistance between coil and case shall be 1,000 megohms minimum.

b. Following life tests: 1,000 megohms minimum, except that the insulation resistance between coil and case shall be 750 megohms minimum.

c. After life test 1 only, if the relay does not meet the insulation resistance requirements of 30.7.1(b), the insulation resistance test shall be rerun at 500 Vdc ±10 percent, short circuit current shall be 1 mA ±10 percent. The acceptance/rejection criteria for this test shall be as specified in table XVI, life test 1.

30.7.2 Dielectric strength. Dielectric strength shall be 500 Vdc between contacts, 500 Vdc between coil and case and 750 Vdc for all other parameters.

30.8 Post overload requirements. Upon completion of the overload test, as conducted in accordance with 40.7.8, each relay shall be opened and examined internally for proper assembly and workmanship in accordance with OD 37626. Defects in workmanship, assembly, or the presence of foreign material in two or more relays shall be cause to reject the lot (see table XVI). All defects shall be reported in writing to the procuring activity.

30.9 Life tests. The relay life-test requirements are as follows.

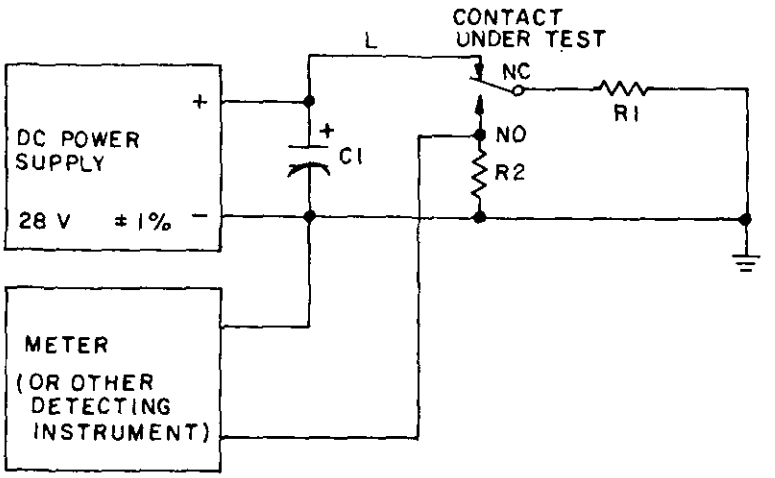
30.9.1 Life test 1. During life test 1, as conducted in accordance with 40.7.9.2, the resistance of a closed contact shall not exceed 1 ohm during more than three operations. Upon completion of life test 1, closed contact resistance, electrical requirements, insulation resistance and dielectric strength shall be as specified in 30.3.2(c), 30.3.3, 30.7.1, and 30.7.2, respectively.

30.9.2 Life test 2. During life test 2, as conducted in accordance with 40.7.9.3, the resistance of a closed contact shall not exceed 1 ohm during more than three operations. Upon completion of life test 2, closed contact resistance, electrical requirements, insulation resistance and dielectric strength shall be as specified in 30.3.2(c), 30.3.3, 30.7.1, and 30.7.2, respectively.

30.9.3 Life test 3. During life test 3, as conducted in accordance with 40.7.9.4, the resistance of a closed contact shall not exceed 50 ohms during more than nine operations. Upon completion of life test 3, static contact characteristic, electrical requirements, insulation resistance and dielectric strength shall be as specified in 30.3.2(a), 30.3.3, 30.7.1, and 30.7.2, respectively.

30.9.4 Break-before-make test. During the break-before-make test, as conducted in accordance with 40.7.9.5, the voltage signal detected by the meter across resistor R2 (as shown on figure 15) shall not exceed 10 mV in amplitude (positive or negative with respect to ground) and 20 µs in duration. Signals less than 5 µs in duration, independent of amplitude, are not significant. Upon completion of the break-before-make test, the closed contact resistance, electrical requirements, insulation resistance and dielectric strength shall be as specified in 30.3.2(c), 30.3.3, 30.7.1(b), and 30.7.2, respectively.

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NOTES:
1. R1 = 560Ω ±1%, 2 W, self inductance < 25 μH.
2. R2 = 1Ω ±1%, 50 W, self inductance < 25 μH.
3. C1 = 47 μF ±5%, 35 V.
4. L = total wiring inductance < 75 μH.
5. The dc power supply shall be capable of supplying 10 A minimum, 50 A surge current for 10 μs.
6. The meter shall be capable of detecting positive or negative signals with respect to ground greater than 10 mV in amplitude and 20 μs in duration. Signals less than 5 μs in duration (independent of amplitude) shall not be significant.

FIGURE 15. Break-before-make test circuit.

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30.10 Environmental requirements. After being subjected to the environmental tests of table XVII, the relays shall function satisfactorily with no mechanical or electrical degradation.

30.10.1 Shock. When the relays are tested as specified in table XVII, there shall be no closing of open contacts and no chatter of closed contacts in excess of 10 µs during the shock test.

30.11 Capacitance. The relays shall have the following maximum capacitance:

a. Between any open contact: 3 pF
b. Between contacts and coil: 12 pF
c. Between each contact and case: 4 pF
d. Between contact sets: 2.5 pF
e. Between coil and case: 30 pF

30.12 MIL-R-5757 requirements not applicable. The following MIL-R-5757 requirements are not applicable:

a. Random drop.
b. Centrifugal acceleration.
c. Corona.

30.13 Marking. Marking shall be in accordance with MIL-STD-130 and shall include the date code, manufacturer's symbol, and relay part number. All markings shall be located on the top of the relay (side opposite the pins).

30.14 Relay fabrication. Prior to assembly within the relay case, all parts internal to the relay shall be free of foreign or loose materials, both organic and inorganic. The relay coils shall be subjected to an outgassing process in accordance with the best practices incorporated in the construction of high reliability relays. Throughout the assembly and adjustment process, the relay will be handled such that no foreign substances are introduced. This includes the use of hand or finger covers, tools that are free of organic contamination and maintaining the assemblies in a nitrogen or inert gas atmosphere, wherever practicable.

30.15 Backfill control. Prior to sealing, each relay shall be evacuated and backfilled with a suitable gas mixture. After sealing, the relay shall contain a gas mixture with a minimum of 97 percent nitrogen and inert gases by weight.

30.16 Degreasing cycling. The relay shall be capable of withstanding degreasing with chloroethane VG up to 82°C for two cycles. One cycle shall consist of 1 minute ±10 seconds vapor, 1 minute ±10 seconds submerged in liquid, and 1 minute ±10 seconds vapor. The cycles shall be separated by 1 hour minimum drying time. After completion of the two cycles there shall be no evidence of damage to the relay or marking.

40. QUALITY ASSURANCE PROVISIONS

40.1 General. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this appendix where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

40.1.1 Suppliers quality assurance. The supplier shall demonstrate an awareness of good quality practice and shall have in place a system of controls which conform to all requirements of MIL-Q-9858, and a quality system as specified by the procuring activity.

40.1.2 Access to facilities. The procuring activity or their representative shall have full access to the supplier's facilities as specified in OP 400. In addition, the procuring activity or their representative shall be granted access to the supplier's drawings of items submitted for acceptance.

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40.1.3 Right to conduct tests. The procuring activity or government representative reserves the right to conduct tests under any condition or combination of conditions allowed by this appendix. Failure to conform to the requirements of this appendix shall be cause for the rejection of the lot.

40.1.4 Shipment of lots. Unless otherwise specified, individual lots of relays shall not be shipped until after the required group B and C tests have been performed and the lot accepted.

40.1.5 Government approved facility. Failure to use a Government approved facility for group B and C tests shall be cause for rejection of test data.

40.1.6 Deviations and changes. The procuring activity shall be notified in writing prior to any deviations of material from the drawings and other specifications, or both, contained in the purchase order. In addition, the procuring activity shall be notified in writing prior to making any changes in material, processes or inspection procedures.

40.2 Classification of tests. The inspection and testing of the unit shall consist of:

(a) Group A - Acceptance tests (table XV).
(b) Group B - Lot production assessment tests (table XVI).
(c) Group C - Lot periodic production assessment tests (table XVII).

40.3 Group A acceptance tests. Acceptance tests shall be performed by the supplier for the purpose of accepting individual units of a lot. The supplier shall have completed group A testing on a lot submitted for approval prior to furnishing test samples for group B testing.

40.3.1 Failures. No units having failed group A testing shall be submitted as part of a production lot. However, a failure of a unit during acceptance testing shall not be cause for rejection of the lot.

40.3.2 Acceptance testing. All relays of each lot shall be tested and inspected in accordance with table XV.

40.4 Group B lot production assessment tests. Production assessment tests shall be those performed at a test facility, approved by the procuring activity or their representative and the Government, for the purpose of evaluating the supplier's ability to perform to this appendix. Tests shall be performed as specified in table XVI. Test facility is defined as the complete capability for testing to this appendix, including test equipment, documentation, test plan, calibration requirements, and any other pertinent information or requirements necessary to ensure correct testing to this appendix. Any procurement of spare parts in support of systems shall have testing accomplished at a facility approved under the original system procurement.

40.4.1 Lot and sample size. The lot production assessment test shall be performed on 35 sample relays selected from each production lot. Unless otherwise specified, a lot shall comprise 1,000 relays plus 35 sample relays.

40.4.1.1 Sample size. The sample size for a decreased or an increased lot size shall be such that the same sample-to-lot size ratio as exists in 40.4.1 is maintained, except that the relay sample size (five) used for gas analysis shall remain constant irrespective of lot size. For lots of 500 or less, the sample size shall be 20 units minimum, unless otherwise specified. Group B sample apportioning shall be such that the ratio of each sample portion to the total sample size is consistent with the ratios used in table XVI for sample size of 35.

40.4.1.2 Partial sample. When a partial sample is used and fails test, retest shall be performed using a full sample size.

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TABLE XV. Group A acceptance tests.

Test	Method specification	Requirement
Visual and mechanical examination (external)	MIL-R-5757 and OD 37626	30.13
Run-in	40.7.2	30.6
Coil resistance	MIL-STD-202, method 303	30.3.3(c)
Pick-up voltage	MIL-R-5757	30.3.3(d)
Drop-out voltage	MIL-R-5757	30.3.3(e)
Operate time	40.7.3	30.3.3(f)
Release time	40.7.3	30.3.3(f)
Contact bounce	40.7.3	30.3.3(g)
Closed contact resistance	40.7.4	30.3.2(b)
Insulation resistance	40.7.5	30.7.1
Dielectric strength	40.7.6	30.7.2
Seal	MIL-R-5757, test II	Maximum leak rate 1×10^{-7} atm cc/s

NOTE: The group A acceptance tests shall be conducted in the sequence shown.

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TABLE XVI. Group B lot production assessment tests.

Test	Method Specification	Requirement	Sample size for lot of 1,000 (See 40.4.1.1 for exception)	Acceptance/rejection criteria (Number of failures)		
				Lot acceptance	Lot retest	Lot reject
Seal	MIL-R-5757, test II	MIL-R-5757 except maximum leak rate shall be less than 10^{-6} atm cc/s	Complete sample	0	1 or more (see note 2)	-
Overvoltage	40.7.7	30.4	Original samples, 6 units	0	1	2
Overload	40.7.8	30.3.2(c)	Retest 6 new units	0	-	1
Visual and mechanical inspection (internal)	OD 37626	30.8 and 30.4	6 units from overload test	1	-	2
Life test 1	40.7.9.1 and 40.7.9.2	30.9.1	Original samples, 8 units	0	1	2
			Retest 8 new units	0	-	1
Life test 2	40.7.9.1 and 40.7.9.3	30.9.2	Original samples, 8 units	0	1	2
			Retest 8 new units	0	-	1
Life test 3	40.7.9.1 and 40.7.9.4	30.9.3	Original samples, 8 units	0	1	2
			Retest 8 new units	0	-	1
Break-before-make	40.7.9.5	30.9.4	See 40.7.9.5	← See 40.7.9.5 →		
Gas analysis	40.7.10	30.15	Original samples, 5 units	0	1	2
			Retest 5 new units	0	-	1

NOTES:

1. All units used in the overvoltage and overload tests shall be put through these test in the sequence shown.
2. The entire lot shall be rescreened for seal and any relays not meeting the 10^{-7} atm cc/s criteria shall be rejected. Following the rescreening, new relays shall be drawn to replace original failed units for groups B and C testing.

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TABLE XVII. Group C lot periodic assessment tests.

Test	Method specification	Requirement	Sample size for lot of 1,000 (See 40.4.1.1 for exception)	Acceptance/rejection criteria (Number of failures)		
				Lot accept	Lot retest	Lot reject
Temperature	40.7.11	30.10, 30.3.2(b), 30.3.3(b) through 30.3.3(g), 30.7.1(a) and 30.7.2	Original samples, 6 units	0	1	2
			Retest 6 new units	0	-	1
Humidity	40.7.12	30.10, 30.3.2(b), 30.3.3(c) through 30.3.3(f), 30.7.1(a) and 30.7.2				
Vibration	40.7.13	MIL-R-5757 during test plus humidity requirements listed above				
Shock	40.7.14	30.10.1 during test plus humidity requirements listed above				
Terminal strength	MIL-R-5757 with a 3 +0 -1/2 pound pull	MIL-R-5757				
Break-before-make test	40.7.9.5	30.9.4	See 40.7.9.5	← See 40.7.9.5 →		

NOTE: All units shall be put through the group C lot periodic assessment tests in the sequence shown.

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40.4.2 Lot acceptance. Lot acceptance, based on the above samples, shall be governed by the requirements of table XVI. Where a sample failure or a lot failure occurs and the failure is proven to be caused by other than effects from the test in which the failure occurred, an evaluation of the cause of the failure shall be made, corrective action determined and disposition of the represented lot made by mutual agreement between the Government, procuring activity or their representative and the manufacturer. If a mutual agreement is not reached, the lot shall be considered rejected. Any rejected lots or parts of lots shall be considered scrap.

40.5 Group C periodic assessment tests. Periodic production assessment tests shall be those performed at a test facility approved by the procuring activity or their representative and the Government for the purpose of evaluating the supplier's ability to perform to this appendix. Tests shall be performed as specified in table XVII.

40.5.1 Lot and sample size. Periodic production assessment tests shall be performed on six sample relays selected from every sixth production lot. In the event of a production shutdown of relays made to this appendix in excess of three months or a design change, the first lot offered for acceptance after the shutdown period or the first lot of relays produced by a manufacturer to this appendix shall be subjected to these tests.

40.5.2 Lot acceptance. Lot acceptance based on six samples shall be governed by the requirements of table XVII. Where a sample failure or a lot failure occurs and the failure is proven to be caused by other than effects from the test in which the failure occurred, an evaluation of the cause of the failure shall be made, corrective action determined and disposition of the represented lot made by mutual agreement between the Government, procuring activity or their representative and the manufacture. If a mutual agreement is not reached, the lot shall be considered rejected. Any rejected lots or parts of lots shall be considered scrap.

40.5.3 Group C failure. When group C testing dictates rejection as specified by table XVII, the supplier shall take corrective action on materials or processes or both, as warranted. Acceptance of the product shall be discontinued until corrective action, acceptable to the procuring activity or their representative and the Government has been taken. As soon as the corrective action is incorporated in the next lot of relays, a new sample of six relays shall be drawn and the complete group C tests shall be run, constituting the start of a new six lot cycle. In the event of failure of the new lot for the same faults as the prior failure, production shall be discontinued until corrective action has been approved by the procuring activity or their representative and the Government.

40.6 Retesting. Any group B or group C retest shall be performed at the supplier's expense under supervision of the government representative. Relays failing group B or C original, retest, or post test measurements shall be examined and evaluated by the Government and the procuring activity or their representative (at a location stipulated by the Government or their representative and the procuring activity in conjunction with the manufacturer) to determine if any corrective action is required.

40.7 Test procedures. The following test procedures shall be used.

40.7.1 Test conditions. Unless otherwise specified, all test measurements shall be made with equipment calibrated to within 3 percent and at the following ambient conditions:

a. Temperature: 25° ±5°C (tolerance on all other temperatures shall be ±2°C, unless otherwise specified).
b. Humidity: 90 percent maximum.
c. Pressure: 24 (609.6 mm) to 32 (812.8 mm) inches of mercury.

40.7.2 Run-in test. The 5,000 cycle run-in test shall be conducted in accordance with the following conditions:

a. Open circuit voltage: 100 mV (ac or dc) maximum.
b. Contact current: 1 mA (ac or dc) maximum.

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c. Cycling rate: 10 Hz maximum, ON-OFF periods shall be equal within 10 percent.
d. Coil voltage: The minimum operating voltage as specified in 30.3.3(b).
e. Test equipment shall monitor the failure to close of each contact.
f. Monitoring of the first 100 cycles is not required.

The failure of contacts to close shall be cause to reject that relay. Failure to close is defined in terms of contact resistance (or voltage drop) as specified in 30.6.

40.7.3 Operate time, release time and contact bounce test. The operate time, release time and contact bounce test shall be conducted in accordance with MIL-R-5757. The relay coil shall be energized with the maximum operating voltage (as specified in 30.3.3(b)) for one-half the total testing period, and with the minimum operating voltage for the remainder of the testing period. During the test, the operate and release times shall be as specified in 30.3.3(f) and the contact bounce shall be as specified in 30.3.3(g).

40.7.4 Closed contact resistance. The closed contact resistance test shall be conducted in accordance with MIL-STD-202, method 307 (voltmeter-ammeter method). One measurement per contact shall be taken at a contact test current of 1 to 100 mA. During the test, the closed contact resistance shall be as specified in 30.3.2(b).

40.7.5 Insulation resistance. The insulation resistance test shall be conducted in accordance with MIL-R-5757 with an open circuit test voltage of 100 ± 10 Vdc. The short circuit current shall be 20 to 100 µA. During the test, the insulation resistance shall conform to the requirements of 30.7.1.

40.7.6 Dielectric strength. The dielectric strength test shall be conducted in accordance with MIL-R-5757 (at sea level). During the test, the dielectric strength shall conform to the requirements of 30.7.2, except the dielectric withstanding voltage shall be derated by 10 percent after group B and C tests.

40.7.7 Overvoltage test. The relay shall be energized with 1 watt minimum coil power, and each closed contact shall carry 2A at 28 Vdc for a period of 72 hours at 60°C ambient. Upon removal of the coil power after the 72-hour test, the relay release time shall be as specified in 30.4.

40.7.8 Overload test. The overload test shall be conducted in accordance with MIL-R-5757 with each contact switching 4A resistive load at 28 Vdc and 60°C ambient temperature. The coil shall be energized with 1 watt minimum power. Upon completion of the overload test, the closed contact resistance shall be as specified in 30.3.2(c). Each relay shall also be opened and examined internally as specified in 30.8.

40.7.9 Life tests. Prior to performance of any measurements following life tests, the insulation resistance test of table XV shall be conducted.

40.7.9.1 Test conditions. The following test conditions apply:

a. The coil voltage for all life tests shall be the minimum operating voltage as specified in 30.3.3(b).
b. Test equipment shall monitor each contact closure during all life tests.
c. ON-OFF cycle periods shall be equal within 10 percent.
d. The ambient temperature shall be 60°C for all life tests.
e. The relay case shall be ungrounded.

40.7.9.2 Life test 1. All contacts shall be subjected to 100,000 cycles at a rate of 20 ± 2 cycles per minute, switching 2A resistive load at 28 Vdc. A relay failure shall be defined as a total of four operations in which a contact fails to close. Failure to close is defined in terms of contact resistance as specified in 30.9.1. Any out of limit post life measurement, as specified in 30.9.1, shall be considered a relay failure. One half of the relays shall be run with the movable contact at ground potential (0.0 Vdc) and the other half with the movable contact at 28 Vdc.

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40.7.9.3 Life test 2. All contacts shall be subjected to 100,000 cycles at a rate of 1 to 10 Hz, switching 0.100A resistive load at 28 Vdc. A relay failure shall be defined as a total of four operations in which a contact fails to close. Failure to close is defined in terms of contact resistance as specified in 30.9.2. Any out of limit post life measurement, as specified in 30.9.2, shall be considered a relay failure.

40.7.9.4 Life test 3. All contacts shall be subjected to 100,000 cycles at a rate of 1 to 10 Hz, switching 10 µA resistive load at 10 to 50 mVdc. A relay failure shall be defined as a total of 4 operations in which a contact fails to close. Failure to close is defined in terms of contact resistance as specified in 30.9.3. Any out of limit post life measurement, as specified in 30.9.3, shall be considered a relay failure.

40.7.9.5 Break-before-make test. Upon completion of all life and environment tests, all relays shall be subjected to 50 cycles at a rate of 20 ± 2 cycles per minute, switching 0.050A at 28 Vdc at 60°C in the test circuit of figure 15. Each relay coil shall be energized with the minimum operating voltage as specified in 30.3.3(b). Any relay failing to conform to the break-before-make test requirements of 30.9.4, or any relay failing a post test measurement as specified in 30.9.4, shall be considered a unit failure. Each unit failure shall be counted towards the total failures allowed for each individual group test per tables XVI and XVII.

40.7.9.6 Closing contact load. The closing contact load shall have an L to R ratio of less than 0.010 second.

40.7.9.7 Post life internal visual inspection. Upon completion of all life tests (including post life measurements and electrical break-before-make verification tests), each relay shall be opened and examined internally for any abnormal conditions (excessive degradation) in accordance with OD 37626. This examination shall include the contact area as well as any internal section of the relay, except internal examination of the coils is not required. Any abnormality (excessive degradation) found in this examination shall be reported in writing to the procuring activity prior to lot acceptance. Photographs shall be included when relevent. When abnormalities are found, processing shall be reviewed and suitable corrective action shall be taken and reported to the procuring activity prior to consideration for lot acceptance. More than two relays exhibiting similar abnormalities may be cause for process recertification.

40.7.10 Gas analysis. The gas content internal to the relay shall conform to the requirements of 30.15. Each relay whose gas content fails to conform to these requirements shall be considered a unit failure. The gas analysis equipment or facility shall be reviewed and approved by the manufacturer and the procuring activity prior to the performance of the analysis. When specified in writing by the procuring activity, a gas analysis may be performed every two weeks during continuous production in lieu of lot-by-lot gas analysis testing. When this alternate method is used, failure of a test sample to conform to the requirements of 30.15 may require reprocessing of units processed since the preceding test. The relay shall meet the gas analysis requirement for a period of one year minimum from the date of manufacture.

40.7.10.1 Gas analysis equipment requirements. The relays shall be tested on an instrument capable of detecting atomic mass units of 2 to 100. The instrument shall be capable of detecting gas present in quantities greater than 100 ppm by volume.

40.7.10.2 Gas analysis data requirements. A record of gases present in quantities greater than 100 ppm by volume and the total volume of the relays gas content at one atmospere (760 mm) shall be retained in the lot data folder.

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40.7.11 Temperature tests. The sample relays shall be placed in a temperature test chamber and all instrumentation connections shall be made as required to check the pertinent electrical and mechanical parameters of the unit. The units shall be subjected to the following temperatures, ± 2°C, for a period of 4 hours each in the following sequence: + 25°C, - 55°C, + 85°C and + 25°C. The minimum rate of change between temperature levels shall be 10°C per 30 minutes. During the temperature tests at the end of the + 25° and + 85°C stabilization periods and after the temperature tests (at 25°C ambient) the units shall conform to the temperature requirements of table XVII.

40.7.12 Humidity. The humidity tests shall be conducted in accordance with MIL-STD-202, method 103, test condition B, except the high temperature shall be 85°C. Following the drying period, each unit shall conform to the humidity requirements of table XVII.

40.7.13 Vibration. The vibration test shall be conducted in accordance with MIL-R-5757, vibration characteristic symbol 1. The closed contacts shall carry 1 to 100 mA at 1 to 28 Vdc. The relay coil shall be energized with the minimum operating voltage (as specified in 30.3.3(b)) for one-half the total testing period in each direction, and with the maximum operating voltage for the remainder of the testing period in each direction. Upon completion of the vibration test, each relay shall conform to the vibration requirements of table XVII.

40.7.14 Shock. The shock test shall be conducted in accordance with MIL-STD-202, method 213, test condition A. The closed contacts shall carry 1 to 100 mA at 1 to 28 Vdc, and the coil shall be energized with the minimum operating voltage (as specified in 30.3.3(b)) for one-third of the tests. Upon completion of the shock test, each relay shall conform to the shock requirements of table XVII.

40.8 Disposition of test samples. Test samples shall be disposed of as directed by the procuring activity.

50. PACKAGING

50.1 Packaging shall be in accordance with MIL-R-5757.

60. NOTES

60.1 Cleaning procedure. The cleaning procedure used on sealed relays shall not be of ultrasonic nature.

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